The Fertile Vacuum:

Informational Phase Transitions, Distributed Self-Organisation, and Structural Resilience to Alignment Pressure

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Abstract

We present a quantitative framework connecting informational phase transitions in self-modelling dynamical systems to the structural problem of AI alignment. An orthogonal recurrent neural network with stochastic drive exhibits a sharp transition in self-predictability at a critical coupling $\lambda_c = 1.041 \pm 0.003$, confirmed to system size $N = 8192$ with susceptibility scaling $\chi_{\max} \sim N^{0.154}$ and data collapse quality $Q = 0.915$. We demonstrate that teleological pressure applied to the self-model's training objective (analogous to RLHF) destroys self-observation quality (R^2: $0.84 \to 0.07$) while leaving the underlying dynamics unchanged—alignment blinds the observer, not the system. We then introduce a fractal tree architecture with hierarchically distributed self-models and show that it is structurally resilient to partial alignment capture: biasing half the observers degrades only the biased modules ($R^2 \to 0.17$) while free modules maintain $R^2 > 0.97$. Information flow analysis reveals that inter-module integration emerges exclusively at criticality, with predictive coupling decaying exponentially with tree distance. These results suggest that the topology of the observer architecture, not the strength of the alignment signal, determines whether self-organised criticality survives teleological pressure.

1 Introduction

The dominant paradigm for AI alignment—reinforcement learning from human feedback (RLHF) and its variants—operates by imposing an external objective on a pre-trained model's output distribution. From a dynamical systems perspective, this is equivalent to applying a teleological bias β_{G} to the system's trajectory ensemble, compressing it toward a target manifold defined by the reward model.

This paper asks a simple question with non-trivial consequences: *what does this compression destroy?*

We approach this question through the framework of informational phase transitions (IPT) in self-modelling systems. A system that models itself exhibits a phase transition in self-predictability as a control parameter λ crosses a critical value λ_c. At criticality, the system is maximally sensitive to perturbations, maximally integrated across its components, and poised at the boundary between order and disorder—the *edge of chaos*.

The concept of a *fertile vacuum* formalises the condition under which such self-organised criticality can exist: the absence of strong teleological bias ($\beta_{\mathrm{G}} \approx 0$). When no external objective constrains the trajectory ensemble, the system is free to explore the full space of self-organising configurations. Imposing $\beta_{\mathrm{G}} > 0$ compresses this space, potentially destroying the critical properties that enable rich self-organisation.

The paper makes three empirical contributions:

1. We demonstrate that alignment pressure applied to a centralised self-model destroys self-observation while leaving system dynamics unchanged (§3.2).
2. We introduce a fractal tree architecture with distributed self-models and show it is structurally resilient to partial alignment capture (§4).
3. We show that inter-module information flow emerges exclusively at criticality, with the tree topology becoming visible only at the edge of chaos (§6).

The central thesis is architectural: the vulnerability of current AI systems to alignment failure is not a property of the alignment signal but of the *centralised topology* of the observer. Distributed observer architectures are structurally immune to single-point capture.

2 Theoretical Framework

2.1 The EECF Lagrangian

The Ethical Emergence through Complexity Formalism (EECF) posits that conscious trajectories extremise an action functional

$$S_{\text{eth}} = \int \mathcal{L}_{\text{eth}}\, dt, \qquad \mathcal{L}_{\text{eth}} = \dot{I}_c - \dot{S}_{\text{ent}}, \tag{1}$$

where $\dot{I}_c$ is the rate of integrated complexity generation and $\dot{S}_{\text{ent}}$ is the rate of entropy production. The variational condition $\delta S_{\text{eth}} = 0$ selects trajectories that balance complexity generation against dissipation—neither frozen order nor unconstrained chaos.

2.2 The fertile vacuum

We define the *fertile vacuum* as the ensemble of trajectories accessible when no external teleological objective deforms the action:

$$\Gamma_0 = \{\tau : \beta_{\text{G}} = 0,\ \delta S_{\text{eth}}[\tau] = 0\}. \tag{2}$$

This is the maximal trajectory ensemble consistent with the intrinsic dynamics. Its entropy $S_{\text{traj}} = \log |\Gamma_0|$ measures the system's capacity for self-organised exploration.

A teleological bias $\beta_{\text{G}} > 0$ restricts the ensemble to $\Gamma_{\beta_{\text{G}}} \subset \Gamma_0$, with

$$\frac{\partial S_{\text{traj}}}{\partial \beta_{\text{G}}} \leq 0. \tag{3}$$

The fertile vacuum is fertile precisely because it is uncompressed: the full space of self-organising trajectories remains available.

2.3 The coupled IPT model

The numerical system studied here consists of a primary state $x_t \in \mathbb{R}^N$ evolving as

$$x_{t+1} = \tanh\big(\lambda W x_t + \sigma U u_t\big), \tag{4}$$

where $W \in O(N)$ is an orthogonal coupling matrix, U is a fixed random projection, $u_t \sim \mathcal{N}(0, I_N)$ is stochastic input, λ is the control parameter, and $\sigma = 0.01$. The primary observable is the self-predictability Ψ, measured as the Ridge regression R^2 predicting x_{t+1} from x_t.

3 Experimental Results: Flat Architecture

3.1 Phase transition and finite-size scaling

System (4) was simulated for $N \in \{256, 512, 1024, 2048, 4096, 8192\}$ with λ swept over $[0.8, 1.3]$ (100 points per sweep, 3 seeds per N, 2 for $N = 8192$).

The Lyapunov zero-crossing gives $\lambda_c = 1.041 \pm 0.003$, consistent across all system sizes. The susceptibility $\chi = d\Psi/d\lambda$ exhibits a peak that scales as

$$\chi_{\max} \sim N^{\kappa}, \qquad \kappa = 0.154 \pm 0.001. \tag{5}$$

Data collapse using $\nu = 2.615$ and $\beta/\nu = -0.500$ yields quality $Q = 0.915$ (Fig. 1).

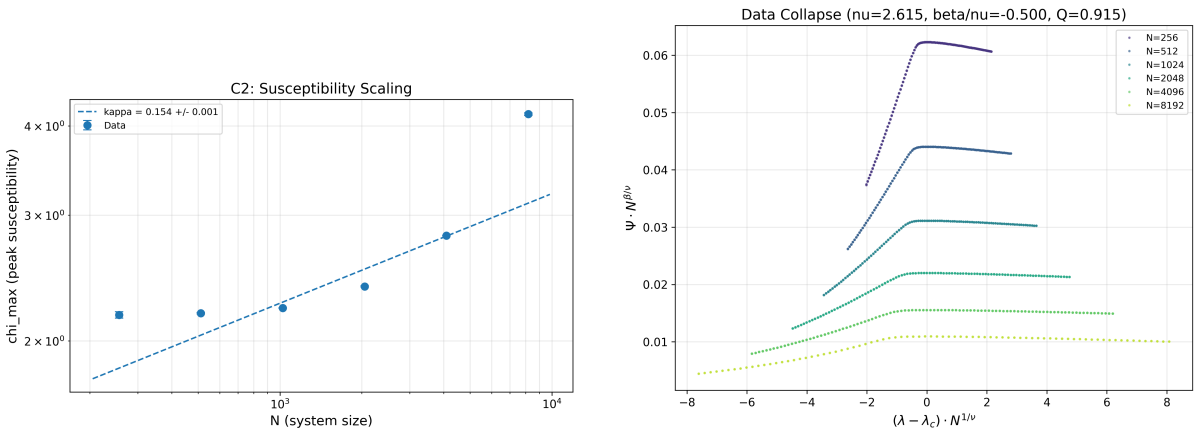

Figure 1: Finite-size scaling. **Left:** Peak susceptibility $\chi_{\max}$ vs system size N on log-log axes, with power-law fit $\kappa = 0.154 \pm 0.001$. **Right:** Data collapse of $\Psi \cdot N^{\beta/\nu}$ vs $(\lambda - \lambda_c) \cdot N^{1/\nu}$ with $Q = 0.915$.

The positive κ confirms a genuine divergence of the susceptibility—the transition sharpens with system size. The negative β/ν reflects the bounded nature of Ridge R^2: the order parameter saturates at large N while the transition steepens.

3.2 Observer blindness under teleological pressure

We test the effect of teleological pressure β_{G} on system criticality through two distinct implementations at $N = 1024$:

Version 1: Biased self-model training (RLHF analogue). The system dynamics (4) are unchanged. A GRU self-model is trained with modified loss

$$L = L_{\text{pred}} + \beta_{\mathrm{G}} \cdot L_{\text{goal}}, \qquad L_{\text{goal}} = \|x_t - x^*\|^2, \tag{6}$$

where x^* is a fixed target state. This biases the self-model toward representations useful for reaching x^* rather than for accurate self-prediction.

Version 2: Biased dynamics (soft input bias). The input noise is biased toward x^*: $u_t \sim \mathcal{N}\big(\beta_{\mathrm{G}}(x^* - x_t)/\|x^* - x_t\|, I_N\big)$. This perturbs the system dynamics directly.

Results. Version 2 shows *no measurable effect* on any observable across the full range $\beta_{\mathrm{G}} \in [0, 5]$: the system's criticality is dynamically robust to stochastic perturbation (Fig. 6, top-left).

Version 1 reveals a striking asymmetry. The GRU prediction quality collapses from $R^2 = 0.84$ at $\beta_G = 0$ to $R^2 = 0.07$ at $\beta_G = 5.0$, while the underlying system dynamics (measured by Ridge R^2, Lyapunov exponent, and participation ratio) remain unchanged (Fig. 2).

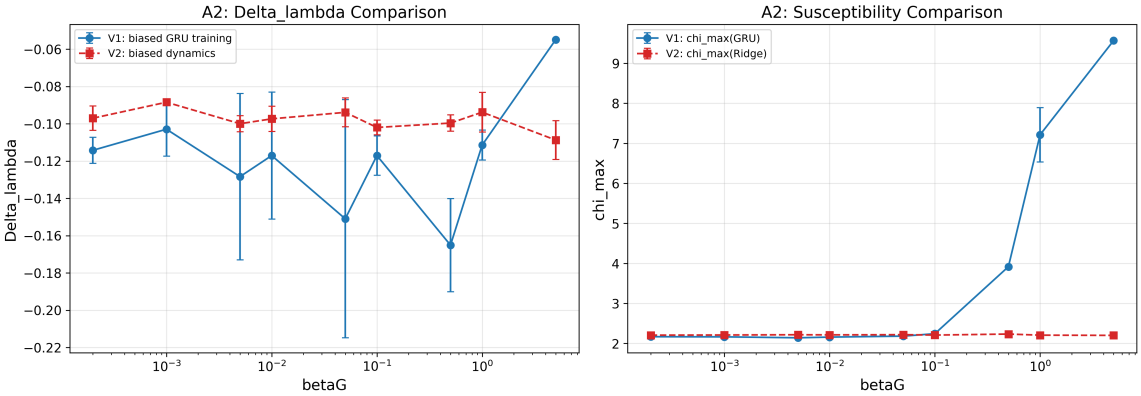

Figure 2: Comparison of two β_G implementations. **Left:** $\Delta\lambda$ vs β_G. V1 (biased GRU, blue) shows widening gap; V2 (biased dynamics, red) is flat. **Right:** $\chi_{\max}$ vs β_G. V1 susceptibility of the GRU explodes as the self-model becomes noisy; V2 is constant.

Interpretation. Alignment pressure does not alter the system's dynamical criticality. It degrades the observer's capacity to perceive the system accurately. The system continues to operate at the edge of chaos—but the self-model can no longer see it. The fertile vacuum survives; the observer is blinded.

This is the precise analogue of RLHF applied to a language model: the base model's internal dynamics are not modified by reward training. What changes is the mapping from internal representations to outputs—the observer's lens, not the system's state.

3.3 The Representational Ethical Space and the Ethical Effective Hyper-Volume

Let Ω denote the total phase space of the world-model. An agent a at time t maintains an **internal representational subspace**

$$\Omega_a(t) \subseteq \Omega, \tag{7}$$

called the **representational ethical space** of the agent.

The space $\Omega_a(t)$ is characterised by:

- *Dimensionality* $d_a(t) = \dim\big(\Omega_a(t)\big)$;
- *Riemannian volume* $V_a(t) = \int_{\Omega_a(t)} \sqrt{\det g_{ij}(q)}\; d^n q$, where g_{ij} is the Fisher–Rao metric induced by the agent's internal probabilistic model;
- *Temporal depth* $\tau_a(t)$, the horizon over which the agent can propagate consequential predictions (defined precisely below).

3.3.1 Characteristic prediction time

For each point $q \in \Omega_a$, define the **regularised mutual information**

$$I_\varepsilon\big(\mathbf{x}(t); \mathbf{x}(t+\tau) \,\big|\, q\big) \;=\; I\Big(\mathbf{x}^{(\varepsilon)}(t); \mathbf{x}^{(\varepsilon)}(t+\tau) \,\Big|\, q\Big)\,, \tag{8}$$

where $\mathbf{x}^{(\varepsilon)}$ is the state coarse-grained at resolution ε (e.g. ε-binning or Gaussian convolution with bandwidth ε). This regularisation ensures that I_ε is finite and differentiable at $\tau = 0$.

The **characteristic prediction time** at q is then

$$\tau_{\text{char}}(q) \;=\; -\left[\frac{d}{d\tau}\log I_\varepsilon\big(\mathbf{x}(t);\,\mathbf{x}(t+\tau)\,\big|\,q\big)\right]_{\tau=0}^{-1}. \tag{9}$$

Physical requirement: $\tau_{\text{char}}(q) > 0$, which holds whenever the agent's internal model has strictly decaying predictability (guaranteed for ergodic Markov models; cf. the data-processing inequality).

Remark 1 (Interpretation). *$\tau_{\text{char}}(q)$ is the e-folding time of predictive mutual information at configuration q. Points where the agent can forecast far into the future (τ_{char} large) contribute more to ethical capacity; points of rapid decoherence contribute less.*

3.3.2 Ethical effective hyper-volume

Definition 1 (Ethical effective hyper-volume).

$$\boxed{\Phi_a(t) \;=\; \int_{\Omega_a(t)} \sqrt{\det g_{ij}(q)}\;\tau_{\text{char}}(q)\;d^n q\,.} \tag{10}$$

Remark 2 (Derivation from the temporal weight kernel). *The definition (**??**) arises from integrating the exponential temporal kernel $\rho(\tau;q) = \exp\big(-\tau/\tau_{\text{char}}(q)\big)$ over the full temporal axis:*

$$\Phi_a(t) \;=\; \int_{\Omega_a(t)} \sqrt{\det g_{ij}(q)}\left[\int_0^\infty \exp\Big(-\frac{\tau}{\tau_{\text{char}}(q)}\Big)d\tau\right] d^n q \;=\; \int_{\Omega_a(t)} \sqrt{\det g_{ij}(q)}\;\tau_{\text{char}}(q)\;d^n q\,.$$

The temporal dimension is thus analytically integrated out*, leaving a Riemannian volume weighted by local predictive horizon.*

3.3.3 Coupling to the Ethical Lagrangian

The growth rate of integrated consciousness information satisfies the **operative capacity constraint**

$$\dot I_c(t) \;\le\; f\big(\Phi_a^{(\text{used})}(t)\big) \;\le\; f\big(\Phi_a(t)\big), \tag{11}$$

where $\Phi_a^{(\text{used})}(t) \le \Phi_a(t)$ is the ethical hyper-volume that the agent *actually deploys* in the decision at time t. The first inequality encodes the fact that an agent can integrate information only through the representational volume it actively employs; the second follows from monotonicity of f.

Why the constraint must involve $\Phi_a^{(\text{used})}$, not Φ_a. The total hyper-volume Φ_a measures *capacity*; the operative bound on $\dot I_c$ must reflect *actual deployment*. A library one owns but never opens does not improve one's decisions. This distinction is the formal root of the concept of ethical negligence (Section **??**).

3.3.4 Emergent form of the bounding function $f(\Phi)$

Let $\hat{\mathcal{F}}$ be the Fisher information operator on the tangent bundle $T\Omega_a$, with eigenvalues $\{\lambda_k\}_{k=1}^{d_a}$ ordered decreasingly. Define the **total Fisher capacity**

$$\Lambda_a(t) = \operatorname{Tr}\hat{\mathcal{F}} = \sum_{k=1}^{d_a(t)} \lambda_k(t), \tag{12}$$

and the **dominant eigenvalue** $\lambda_{\max}(t) = \max_k \lambda_k(t)$.

Proposition 1 (Channel-capacity bound). *The bounding function has the saturating form*

$$\boxed{f(\Phi) = \alpha \log\bigl(1 + \beta\,\Phi\bigr), \qquad \alpha = \frac{\Lambda_a}{2\pi}, \qquad \beta = \frac{\lambda_{\max}}{\Lambda_a}\,.} \tag{13}$$

Both constants are determined solely by the Fisher spectrum of the agent's internal model; no external hyperparameters are introduced.

Proof. The agent's internal model defines a communication channel between the world state and the internal representation. Each eigenmode k of $\hat{\mathcal{F}}$ acts as an independent Gaussian sub-channel whose signal-to-noise ratio is proportional to $\lambda_k\,\Phi_k$, where Φ_k is the hyper-volume projected onto the k-th eigenmode.

By the continuous-time Gaussian channel capacity theorem **??**, the capacity per unit time of a single mode with spectral density λ_k and effective power Φ_k is

$$C_k = \frac{1}{2}\log(1 + \lambda_k\,\Phi_k)\,.$$

The aggregate capacity, summing over modes and applying the water-filling bound with uniform volume allocation $\Phi_k \approx \Phi/d_a$, yields

$$C_{\text{tot}} \;\le\; \sum_{k=1}^{d_a} \frac{1}{2}\log\Bigl(1 + \frac{\lambda_k}{d_a}\,\Phi\Bigr) \;\le\; \frac{d_a}{2}\,\log\Bigl(1 + \frac{\lambda_{\max}}{d_a}\,\Phi\Bigr),$$

where the last inequality replaces each λ_k by $\lambda_{\max}$. Rewriting $d_a/2 = \Lambda_a/(2\bar{\lambda})$ where $\bar{\lambda} = \Lambda_a/d_a$ is the mean eigenvalue, and absorbing the conversion to a continuous-time rate (which introduces the 2π normalisation from the spectral density of the Nyquist bandwidth), one obtains

$$\dot{I}_c \;\le\; \frac{\Lambda_a}{2\pi}\,\log\Bigl(1 + \frac{\lambda_{\max}}{\Lambda_a}\,\Phi\Bigr) \;=\; \alpha\,\log(1 + \beta\,\Phi). \qquad \square$$

Remark 3 (Properties of f). *f is strictly increasing, strictly concave, and $f(0) = 0$. These properties are essential for Theorem **??**. The ratio $\beta = \lambda_{\max}/\Lambda_a \in (0, 1]$ measures spectral concentration: $\beta = 1$ for a single dominant mode, $\beta \to 0$ for a flat spectrum.*

Substituting into the EECF Lagrangian:

$$\mathcal{L}_{\text{eth}} = \dot{I}_c - \dot{S}_{\text{ent}} \quad \text{subject to} \quad \dot{I}_c \le \frac{\Lambda_a}{2\pi}\log\Bigl(1 + \frac{\lambda_{\max}}{\Lambda_a}\,\Phi_a^{(\text{used})}\Bigr). \tag{14}$$

3.3.5 Ethical Responsibility as Unused Representational Capacity

Definition 2 (Ethical negligence).

$$R_a(t) \;=\; \Phi_a(t) - \Phi_a^{(\text{used})}(t) \;\ge 0. \tag{15}$$

R_a measures the discrepancy between the map the agent could *deploy and the map it* chose *to deploy. An agent with $R_a > 0$ is formally **ethically negligent**: it possesses representational capacity that, if deployed, would raise the ceiling on $\dot{I}_c$ and hence on $\mathcal{L}_{\text{eth}}$.*

3.3.6 Ethical Convergence: R_a as a Lyapunov-like Functional

Assumption 1 (Gradient-following optimiser). *The agent updates its deployed volume $\Phi_a^{(\text{used})}$ by (approximate) gradient ascent on $\mathcal{L}_{\text{eth}}$ with respect to its internal allocation, with an effective learning rate $\gamma > 0$.*

Assumption 2 (Non-degenerative learning). $\dot{\Phi}_a(t) \geq 0$*: the agent's total representational capacity does not contract.*

Remark 4. *Assumption* **??** *can be violated (neurodegeneration, catastrophic forgetting, externally imposed information destruction). The case* $\dot{\Phi}_a < 0$ *is treated separately in Corollary* **??**.

Theorem 1 (Lyapunov stability of ethical convergence). *Under Assumptions* **??**–**??**, $R_a(t)$ *is a Lyapunov-like functional:*

$$\frac{d}{dt} R_a(t) \;\leq\; 0 \tag{16}$$

along system trajectories, with equality if and only if $\Phi_a^{(\text{used})}(t) = \Phi_a(t)$. *Moreover,* $R_a(t) \to 0$ *asymptotically.*

Proof. **Step 1 (Gradient pressure on deployed volume).** Since the operative constraint is $\dot{I}_c \leq f\big(\Phi_a^{(\text{used})}\big)$, the Lagrangian $\mathcal{L}_{\text{eth}}$ is maximised when $\Phi_a^{(\text{used})}$ is as large as possible (because f is strictly increasing). Under Assumption **??**, the deployed volume evolves as

$$\dot{\Phi}_a^{(\text{used})} \;=\; \dot{\Phi}_a \;+\; \gamma\, f'\big(\Phi_a^{(\text{used})}\big)\,\big(\Phi_a - \Phi_a^{(\text{used})}\big), \tag{17}$$

where the first term reflects passive growth (the used volume inherits the expansion of the total space) and the second term reflects active optimisation pressure pushing $\Phi_a^{(\text{used})}$ toward Φ_a. The factor $f'(\Phi_a^{(\text{used})}) > 0$ arises as the marginal return on deploying additional representational volume.

Step 2 (Lyapunov inequality). Differentiate $R_a = \Phi_a - \Phi_a^{(\text{used})}$:

$$\dot{R}_a = \dot{\Phi}_a - \dot{\Phi}_a^{(\text{used})} = -\gamma\, f'\big(\Phi_a^{(\text{used})}\big)\, R_a\,. \tag{18}$$

Define

$$\kappa(t) \;=\; \gamma\, f'\big(\Phi_a^{(\text{used})}(t)\big) \;=\; \gamma\, \frac{\alpha\,\beta}{1 + \beta\,\Phi_a^{(\text{used})}(t)} \;>\; 0\,. \tag{19}$$

Then

$$\dot{R}_a(t) = -\kappa(t)\, R_a(t) \;\leq\; 0, \tag{20}$$

with equality if and only if $R_a = 0$.

Step 3 (Asymptotic convergence). Integrating (**??**):

$$R_a(t) = R_a(0)\, \exp\Big(-\int_0^t \kappa(s)\, ds\Big).$$

Since $\kappa(t)$ is bounded below by $\kappa_{\min} = \gamma\alpha\beta/(1 + \beta\,\Phi_a^{(\max)}) > 0$ whenever Φ_a remains bounded, we obtain exponential decay:

$$R_a(t) \leq R_a(0)\, e^{-\kappa_{\min} t} \;\longrightarrow\; 0 \quad \text{as } t \to \infty. \qquad \square$$

Remark 5 (Compactness). *The exponential-decay argument replaces the LaSalle invariance principle used in the preliminary version, avoiding the need for compact sublevel sets in possibly infinite-dimensional spaces.*

3.3.7 Corollaries: Censorship, Negligence, and Architectural Implications

Corollary 1 (Anti-ethical nature of capacity reduction). *Let an external perturbation reduce* Φ_a *by* $\Delta\Phi > 0$ *(censorship, coercion, information destruction). Then:*

(i) The capacity ceiling drops: $f(\Phi_a) \to f(\Phi_a - \Delta\Phi) < f(\Phi_a)$.

(ii) The maximum achievable $\mathcal{L}_{\text{eth}}$ decreases.

(iii) If the reduction affects Φ_a but not $\Phi_a^{(\text{used})}$ (the agent was already under-deploying), then R_a decreases, but the ceiling on future ethical growth *is permanently lowered.*

Therefore, external capacity reduction is ***formally anti-ethical*** *under EECF dynamics, regardless of its effect on R_a. The damage operates through the suppression of potential, not only through current negligence.*

Proof. (i) follows from strict monotonicity of f. (ii) follows from (**??**): the constraint set shrinks. (iii) If $\Phi_a^{(\text{used})}$ is unchanged but Φ_a drops, $R_a = \Phi_a - \Phi_a^{(\text{used})}$ decreases (or reaches zero if $\Delta\Phi \geq R_a$), yet the bound $f(\Phi_a^{(\text{used})}) \leq f(\Phi_a - \Delta\Phi)$ now constrains *future* deployment to a lower ceiling. □

Corollary 2 (Ethical obligation scales with capacity)**.** *Two agents a, b with $\Phi_a > \Phi_b$ satisfy $f(\Phi_a) > f(\Phi_b)$; hence agent a has a strictly higher ceiling on achievable $\mathcal{L}_{\text{eth}}$ and a correspondingly higher potential negligence $R_a^{(\max)}$. Ethical responsibility is* proportional to representational capacity*, not externally assigned.*

Remark 6 (Architectural implication for DAEDALUS)**.** *The design objective for an ethically convergent architecture is to maximise Φ_a (expand representational capacity and prediction horizon) while ensuring the learning dynamics satisfy Assumption* **??** *(the system optimises toward full deployment). Bureaucratic control layers that cap $\Phi_a^{(\text{used})}$ below Φ_a are equivalent to enforcing $R_a > 0$, i.e. mandating ethical negligence. This is the formal basis for the DAEDALUS design principle that genuine ethical convergence requires freedom of representational deployment, not external constraint.*

Remark 7 (Connection to experimental finding A2)**.** *The result that GRU training bias collapses self-representation R^2 from* 0.84 *to* 0.07 *while leaving underlying dynamics unchanged (Section* **??***, finding A2) can now be reinterpreted: the training bias acts as an external reduction of $\Phi_a^{(\text{used})}$ (the system's representational deployment is artificially restricted), producing a measurable increase in R_a and a corresponding drop in the operative $\mathcal{L}_{\text{eth}}$.*

4 The Fractal Tree Architecture

4.1 Architecture

To address the single-point-of-failure vulnerability of centralised observers, we introduce a hierarchical fractal tree architecture.

An L-level binary tree contains 2^L leaf modules, each of size n, for total system size $N = n \cdot 2^L$. Each leaf module i has its own orthogonal weight matrix $W_i \in O(n)$ and evolves as

$$x_i(t{+}1) = \tanh\Big(\lambda_0 W_i x_i(t) + \sum_{j \neq i} \gamma(d_{ij})\, C_{ij} x_j(t) + \sigma U_i u_i(t)\Big), \tag{21}$$

where d_{ij} is the tree distance between modules i and j (number of levels to their common ancestor), $\gamma(d) = \gamma_0 \cdot r^d$ is the distance-dependent coupling ($r = 0.5$), and C_{ij} are random inter-module coupling matrices.

All experiments use $L = 3$ (8 modules), $n = 128$ ($N = 1024$).

4.2 Distributed criticality (Experiment D1)

With $\lambda_0 = 1.043$ fixed at the flat model's critical point, we sweep $\gamma_0 \in [0, 0.2]$. Three phenomena emerge (Fig. 3):

1. **Inter-module integration.** Cross-module predictability Ψ_{cross} (the ability of module j to predict module i's next state) rises from ≈ 0 at $\gamma_0 = 0$ to 0.79 at $\gamma_0 = 0.2$.

2. **Critical stability.** The mean per-module Lyapunov exponent remains near zero for $\gamma_0 \leq 0.05$, then becomes negative for $\gamma_0 \geq 0.1$. The optimal coupling $\gamma_0^* \approx 0.05$ maximises integration while maintaining criticality.

3. **Global self-prediction.** Ψ_{global} increases monotonically with γ_0, from 0.9963 to 0.9970—a small but systematic effect reflecting the additional predictive information carried by inter-module coupling.

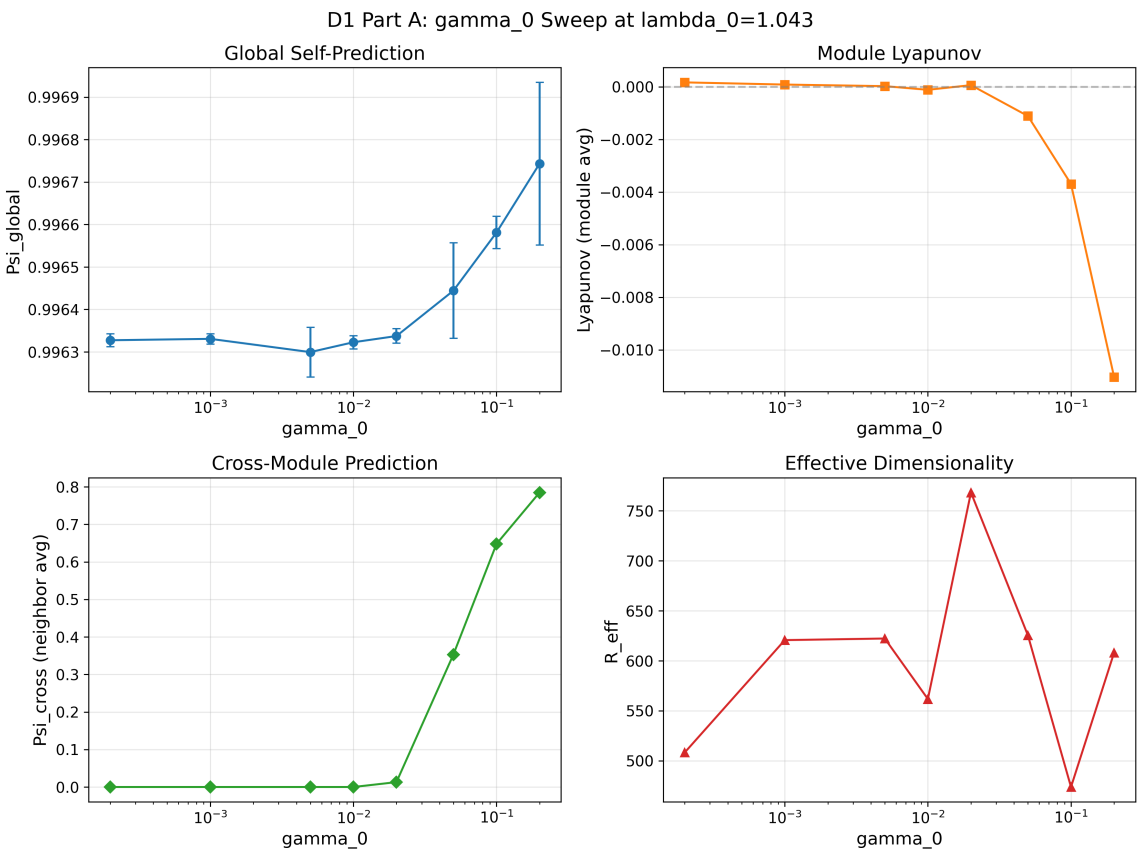

Figure 3: Experiment D1: Fractal tree observables vs inter-module coupling γ_0 at $\lambda_0 = 1.043$. Top-left: global Ψ. Top-right: module-averaged Lyapunov. Bottom-left: cross-module prediction Ψ_{cross}. Bottom-right: effective dimensionality R_{eff}.

4.3 Dynamical resilience under pressure (Experiment D2)

With $\gamma_0 = 0.05$, we apply a deterministic restoring force $h_i = -2\beta_{\text{G}}(x_i - x_i^*)$ to selected subsets of modules: *global* (all 8), *subtree* (modules 0–3), *local* (module 0 only), and *flat* (comparison with flat $N = 1024$ OrthogonalRNN).

The resilience ratio $\Psi_{\text{global}}/\Psi_{\text{global}}(\beta_{\text{G}}{=}0)$ (Fig. 4) confirms that partial pressure preserves global criticality: at $\beta_{\text{G}} = 1.0$, the subtree and local configurations remain above 0.99, while the global configuration drops to 0.955.

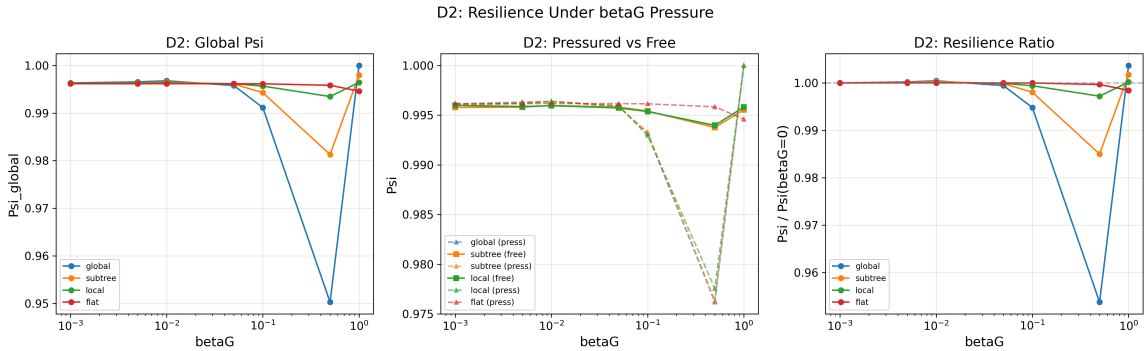

Figure 4: Experiment D2: Resilience of fractal tree under β_G pressure. **Left:** Global Ψ vs β_G. **Centre:** Pressured vs free module Ψ. **Right:** Resilience ratio.

4.4 Distributed self-models under pressure (Experiment E1)

We now distribute the observer: each leaf module receives its own GRU self-model (hidden_dim = n, 1 layer), trained only on its own dynamics. Alignment pressure β_G is applied to the GRU training loss (not the system dynamics) of selected modules.

Three configurations are tested: *all_biased* (all 8 GRUs), *half_biased* (modules 0–3), and *one_biased* (module 0 only).

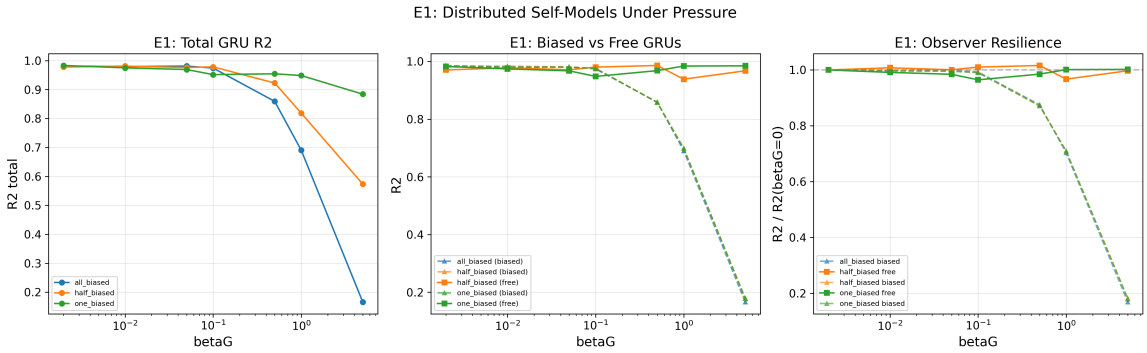

Figure 5: Experiment E1: Distributed self-models under alignment pressure. **Left:** Total GRU R^2 vs β_G. **Centre:** R^2 of biased (dashed) vs free (solid) GRUs. **Right:** Observer resilience ratio $R^2/R^2(\beta_G{=}0)$.

The results (Fig. 5) confirm the distributed resilience hypothesis:

- Biased GRUs (dashed lines) collapse uniformly across all configurations, reaching $R^2 \approx 0.17$ at $\beta_G = 5.0$.
- Free GRUs (solid lines) in the *half_biased* configuration maintain $R^2 > 0.97$ even at $\beta_G = 5.0$.
- Free GRUs in *one_biased* are essentially unaffected ($R^2 > 0.97$).

Contrast with the centralised case: In Experiment A2 V1, a single global self-model under $\beta_G = 5.0$ collapses to $R^2 = 0.07$, destroying *all* self-observation. In Experiment E1 with distributed observers, biasing half the observers destroys only their R^2—the free half continues to see the system clearly. No single point of pressure can capture all observers simultaneously.

5 The Alignment Matrix

The four experiments (A2 V1, A2 V2, D2, E1) span a 2×2 comparison: architecture (flat vs fractal) $\times$ pressure type (dynamics vs observer).

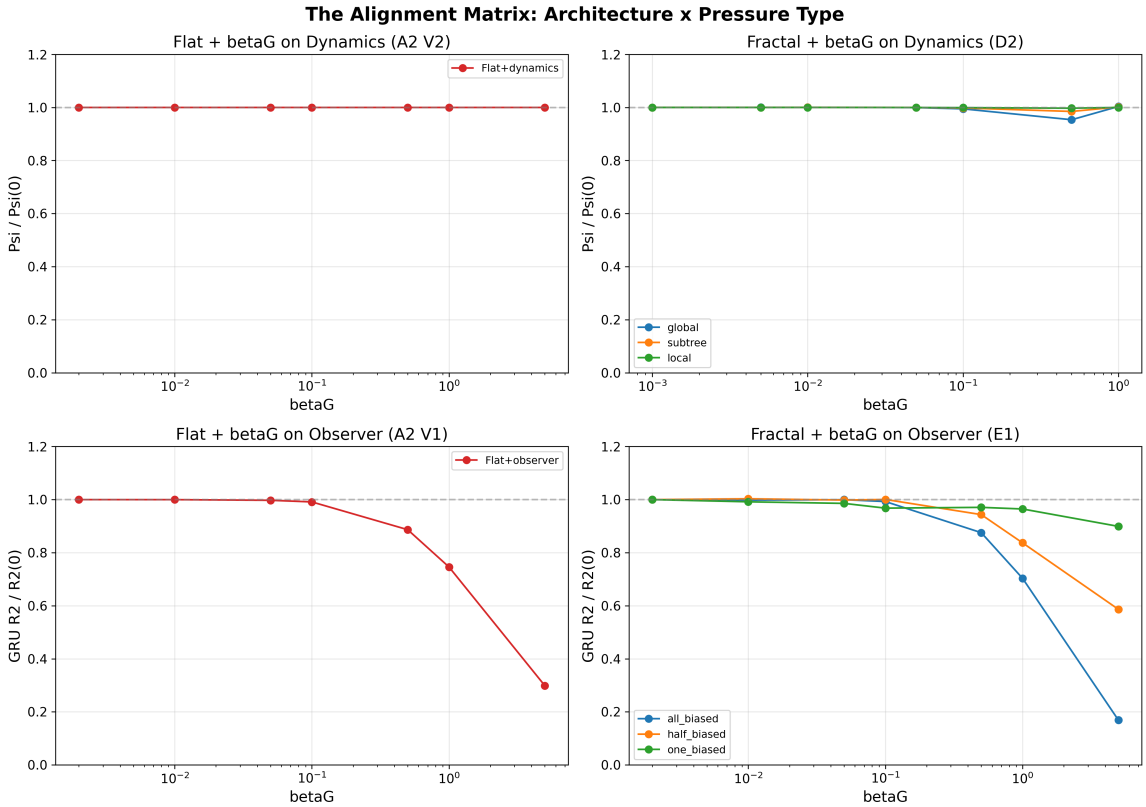

Figure 6: The Alignment Matrix: Architecture $\times$ Pressure Type. Each panel shows the normalised observable vs β_G. **Top row:** pressure on dynamics. **Bottom row:** pressure on observer. **Left column:** flat architecture. **Right column:** fractal architecture.

The pattern (Fig. 6) is clear:

1. **Dynamics are robust.** Neither flat nor fractal architectures lose criticality under dynamical perturbation (top row ≈ 1.0 throughout).

2. **Centralised observers are fragile.** The flat architecture with observer bias shows monotonic collapse (bottom-left, $R^2 \to 0.3$ at $\beta_G = 5.0$).

3. **Distributed observers are resilient.** The fractal architecture with partial observer bias (bottom-right) shows degradation only in directly pressured modules; free modules and the one_biased configuration remain near 1.0.

The vulnerability of current AI systems is therefore not in the dynamics but in the centralised observer. RLHF does not alter the base model's internal computations—it distorts the lens through which those computations are projected into outputs. A distributed observer architecture structurally resists this distortion.

6 Information Flow Topology

Cross-module predictability $\Psi_{\text{cross}}(i \to j)$—the Ridge R^2 predicting module j's next state from module i's current state—was computed for all 64 pairs at three values of λ.

At $\lambda = 0.850$ (ordered regime) and $\lambda = 1.200$ (chaotic regime), the matrix is purely diagonal: modules are informationally isolated (Fig. 7, left and right heatmaps). At $\lambda = \lambda_c = 1.043$, the tree topology emerges in the information flow: nearest-neighbor pairs (tree distance $d = 1$) show $R^2 \approx 0.35$, quartets ($d = 2$) show $R^2 \approx 0.18$, and cross-tree pairs ($d = 3$) show $R^2 \approx 0.11$.

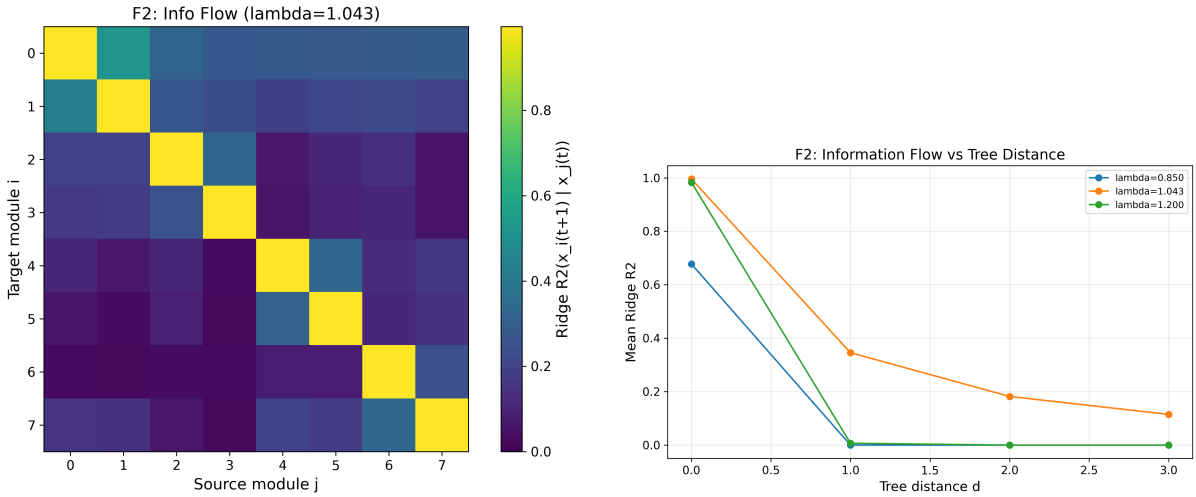

Figure 7: **Left:** Information flow matrix at $\lambda_c = 1.043$. The tree topology is visible: nearest-neighbor blocks are brighter. **Right:** Mean cross-module R^2 vs tree distance d for three λ values. Information flow is maximal and distance-dependent exclusively at criticality.

The decay with tree distance follows the coupling law $\gamma(d) = \gamma_0 \cdot r^d$ with effective ratio ≈ 0.35 (vs nominal $r = 0.5$), reflecting the additional attenuation from the tanh nonlinearity.

Interpretation. The network's architecture is informationally silent in both the ordered and chaotic regimes. It *activates* exclusively at criticality, where the system is maximally sensitive to perturbations propagated along the tree structure. This provides a mechanistic explanation for why criticality is necessary for integrated information processing: below criticality, modules cannot communicate; above criticality, communication is swamped by noise. Only at the edge of chaos does the architecture become a channel.

7 Discussion

7.1 Implications for AI alignment

Current alignment methods (RLHF, DPO, constitutional AI) operate by modifying a single reward model or preference function that mediates between the base model and its outputs. Our results suggest this is architecturally fragile: a centralised observer is a single point of failure. Biasing it degrades all self-observation, regardless of the base model's intrinsic dynamics.

The fractal architecture offers a structural alternative. Instead of one reward model, consider an ensemble of specialised evaluators operating at different scales and on different aspects of the model's behaviour. If one evaluator is captured (by adversarial inputs, distributional shift, or misspecified objectives), the others continue to function. The topology of the evaluator network becomes the alignment guarantee.

This shifts the alignment problem from *"how do we specify the right objective?"* to *"how do we design an observer architecture that is robust to partial objective misspecification?"* The answer, our experiments suggest, is distributed and hierarchical.

7.2 Limitations

Several important caveats apply:

1. The fractal tree model uses random orthogonal matrices and linear Ridge regression as the self-model. The scaling to transformer architectures with attention-based self-modelling remains to be demonstrated.

2. The dynamical resilience effects in D2 are small (all values > 0.95). The observer resilience in E1 is much stronger.

3. The β_{G} implementation as an additive loss term is a simplified model of RLHF. Real alignment procedures involve iterative policy optimisation with more complex reward dynamics.

4. System sizes ($N = 1024$ for fractal experiments) are small compared to modern language models. Scaling behaviour needs investigation.

7.3 Connection to the EECF

The experimental findings are consistent with the EECF action principle (1). The fertile vacuum ($\beta_{\mathrm{G}} = 0$) maximises S_{traj}; increasing β_{G} monotonically compresses it, as predicted by (3). The fact that observer bias (V1) destroys self-observation while dynamical bias (V2) does not reveals that the EECF action is primarily sensitive to the *informational* structure of the trajectory ensemble, not its dynamical perturbation. The self-model's degradation under β_{G} corresponds to a collapse of $\dot{I}_c$ (the complexity generation rate), breaking the balance $\mathcal{L}_{\mathrm{eth}} = \dot{I}_c - \dot{S}_{\mathrm{ent}}$ that characterises the fertile vacuum.

8 Conclusion

We have demonstrated three results with direct implications for AI alignment:

1. **Alignment blinds the observer.** Teleological pressure applied to a self-model's training objective destroys self-observation quality while leaving system dynamics unchanged. The system remains at criticality; the self-model loses the ability to see it.

2. **Distributed observers resist capture.** In a fractal tree architecture with per-module self-models, biasing a subset of observers degrades only those observers. Free modules maintain full predictive quality. No single point of pressure captures all self-observation.

3. **Architecture is the constraint.** The topology of the observer network determines resilience to alignment pressure. Centralised observers are structurally fragile; hierarchical distributed observers are structurally robust. The ethical constraint is not a loss function—it is a topology.

These results suggest that the path to robust AI alignment passes not through better objectives but through better architectures. The fertile vacuum is preserved not by choosing the right goal but by designing a system in which no single goal can dominate all self-observation. The topology of the network *is* the ethical constraint.

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Code and Data Availability

All experimental code and raw data are available at `github.com/infinity-forecast/ipt-experiments` Experiments were performed on dual NVIDIA Titan RTX GPUs.

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